RIZ MUGHAL



QUIZ MASTER

MTH603 QUIZ 2 (Lecture 23 to 28)

100% correct solution.

For more information, you can visit my channel and for any type of help related to CS619 you can contact me.

YOUTUBE CHANNEL:

https://www.youtube.com/channel/UCINsFwDiB62SValCcPDZbRQ/playlists

FACEBOOK GROUP:

https://www.facebook.com/groups/923887914750307

MTH603:Quiz no.2 Quiz Start Time: 08:40 AM, 30 Janua

Question # 1 of 10 (Start time: 08:40:29 AM, 30 January 2021) Total M

For the given data points $(2, 5)$, $(4, 7)$, *and* $(6, 9)$, *the zero − order divide difference will be*

Select the correct option

Reload Math Equa

- ○ 0
- ○ 1
- ○ 2
- ◉ 5

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MTH603:Quiz no.2 Quiz Start Time: 08

Question # 2 of 10 (Start time: 08:40:59 AM, 30 January 2021)

For the following data

x	1	2	5	7	8
y	-5	10	20	22	24

the polynomial of the Lagrangens interpolation could be

Select the correct option

- $\frac{4}{41}x^7 - \frac{43}{7}x^2 + \frac{65}{28}x - \frac{186}{5}$
- $-\frac{11}{280}x^4 + \frac{419}{420}x^3 - \frac{7603}{840}x^2 + \frac{15019}{420}x - \frac{98}{3}$
- $-\frac{1121}{80}x^5 + \frac{41}{7}x^3 - \frac{163}{40}x^2 + \frac{19}{81}x - \frac{3}{10}$
- $x^6 - \frac{1}{56}x^5 + \frac{47}{5}x^4 - \frac{67}{90}x + \frac{2}{5}$

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MTH603:Quiz no.2 Quiz Start Time: 08:40 AM, 30 Janu

Question # 3 of 10 (Start time: 08:41:28 AM, 30 January 2021) Total

If only two data points *are given, the formula for Lagrange's* interpolation *polynomial will be*

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Select the correct option

Reload Math Equ

- $y = f(x) = \frac{(x - x_0)}{(x_1 - x_0)} y_0 + \frac{(x - x_1)}{(x_0 - x_1)} y_1$
- (selected) $y = f(x) = \frac{(x - x_1)}{(x_0 - x_1)} y_0 + \frac{(x - x_0)}{(x_1 - x_0)} y_1$
- $y = f(x) = \frac{(x - x_0)}{(x_0 - x_1)} y_0 + \frac{(x - x_1)}{(x_1 - x_0)} y_1$
- $y = f(x) = \frac{(x_1 - x_0)}{(x - x_0)} y_0 + \frac{(x_0 - x_1)}{(x - x_1)} y_1$

MTH603:Quiz no.2 Quiz Start Time:

Question # 4 of 10 (Start time: 08:41:51 AM, 30 January 2021)

If y(x) is approximated by a polynomial $P_n(x)$ of degree n then the error is given by

Select the correct option

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- ○ $\varepsilon(x) = y(x) + P_n(x)$
- ◉ $\varepsilon(x) = y(x) - P_n(x)$
- ○ $\varepsilon(x) = y(x) \times P_n(x)$
- ○ $\varepsilon(x) = y(x) \div P_n(x)$

MTH603:Quiz no.2 Quiz Sta

Question # 5 of 10 (Start time: 08:42:13 AM, 30 January 2021)

If any ten data points are given,the degree of Lagrange's interpolation polynomial could be

Select the correct option

- (●) nine
- ( ) ten
- ( ) eleven
- ( ) twelve

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Question # 6 of 10 (Start time: 08:42:36 AM, 30 January 2021)

Given the following data x: 0 1 2 4 y: 1 1 2 6 Value of 1st order divided difference f[2,4] is

Select the correct option

- ○ 1
- ◉ 2
- ○ 0
- ○ -2

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MTH603:Quiz no.2 Quiz Start Time: 08:40 AM, 30 Janua

Question # 7 of 10 (Start time: 08:43:04 AM, 30 January 2021) Total

For the given data points $(2, 0.3)$, $(4, 1)$, *and* $(6, 1.2)$ *the divide difference table will be given as*

Select the correct option

Reload Math Equ

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	x	y	$1stD.D$	$2ndD.D$
	2	0.3	-0.0625	
	4	1	0.1	0.35

	x	y	$1stD.D$	$2ndD.D$
	2	0.3	0.35	
	4	1	0.1	-0.0625
	6	1.2		

	x	y	$1stD.D$	$2ndD.D$
	2	0.3	-0.0625	
	4	1	0.35	0.1

	x	y	$1stD.D$	$2ndD.D$
	2	0.3	0.1	
	4	1	0.0625	0.25

MTH603:Quiz no.2 Quiz

Question # 8 of 10 (Start time: 08:43:34 AM, 30 January 2021)

Given the following data x:0 1 4 y:2 1 4 Value of first order divided difference y[0,1]is

Select the correct option

○	-2
○	1
◉	-1
○	2

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sec(s)

MTH603:Quiz no.2 Quiz Start Time: 08:40 AM, 30 January 2021

Question # 9 of 10 (Start time: 08:44:01 AM, 30 January 2021) **Total Marks:** 1

If there are (n+2) values of y corresponding to (n+2) values of x, then we can represent the function f(x) by a polynomial of degree

Select the correct option

Reload Math Equations

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- ○ n+2
- ◉ n+1
- ○ n
- ○ n-1

sec(s)

MTH603:Quiz no.2 Quiz Start Time: 08:40 AM, 30 January 2021

Question # 10 of 10 (Start time: 08:44:33 AM, 30 January 2021) Total Marks: 1

For the giev three data points, *the* degree *of Lagrange's* interpolation *polynomial could be*

x	0.3	0.7	0.9
y	0.067	0.248	0.518

Select the correct option

Reload Math Equations

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- $y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.\ 7)(0.3-0.9)}(0.248) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.067) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.518)$
- $y = f(\ = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.518) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.248) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.067)$
- $y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.067) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.248) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.518)$
- $y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.067) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.518) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.248)$

2nd account

MTH603:Quiz no.2 Quiz Start

Question # 1 of 10 (Start time: 08:48:44 AM, 30 January 2021)

If y(x) is approximated by a polynomial $P_n(x)$ of degree n then the error is given by

Select the correct option

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- ○ $\varepsilon(x) = y(x) + P_n(x)$
- ● $\varepsilon(x) = y(x) - P_n(x)$
- ○ $\varepsilon(x) = y(x) \times P_n(x)$
- ○ $\varepsilon(x) = y(x) \div P_n(x)$

MTH603:Quiz no.2 Quiz Start Time: 08:48 AM, 30 Januar

Question # 2 of 10 (Start time: 08:49:11 AM, 30 January 2021) Total M

For the given data points $(4, 1.3)$, $(8, 1.5)$, *and* $(12, 1.9)$ *the divide difference table will be given as*

Select the correct option

Reload Math Equa

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x	y	$1stD.\ D$	$2ndD.\ D$
4	1.3	0.0062	
8	1.5	0.1	0.05

x	y	$1stD.\ D$	$2ndD.\ D$
4	1.3	0.05	
8	1.5	0.1	0.0062
12	1.9		

x	y	$1stD.\ D$	$2ndD.\ D$
4	1.3	0.1	
8	1.5	0.35	0.0062

x	y	$1stD.\ D$	$2ndD.\ D$
4	1.3	0.1	

MTH603:Quiz no.2

Quiz Start Time: 08:48 AM, 30 Janua

Question # 3 of 10 (Start time: 08:49:37 AM, 30 January 2021)

Total

Which of the following method can be used for interpolation *for the given values of* x *and* y*?*

x	0.3	0.7	0.9
y	0.067	0.248	0.518

Select the correct option

Reload Math Equ

- ○ Newton's forward difference formula
- ○ Newton's backward difference formula
- ○ Newton's interpolation form
- ● Lagrange's interpolation formula

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MTH603:Quiz no.2 Quiz Start Time: 08:48 AM, 30 Ja

Question # 4 of 10 (Start time: 08:50:05 AM, 30 January 2021) To

For the given four data points, *the* degree *of Lagrange's* interpolation *polynomial could be*

x	0.3	0.7	0.9	1.0
y	0.067	0.248	0.518	0.6812

Select the correct option

Reload Math

- six
- five
- four
- three

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MTH603:Quiz no.2 Quiz Start Time: 08:48 AM, 30 January 2021

Question # 5 of 10 (Start time: 08:50:33 AM, 30 January 2021) Total Marks: 1

If any three data points *are given, the formula for Lagrange's* interpolation *polynomial will be*

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Select the correct option

Reload Math Equations

- $y = f(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_0$
- $y = f(x) = \frac{(x_0-x_1)(x_0-x_2)}{(x-x_0)(x-x_2)}y_0 + \frac{(x_1-x_0)(x_1-x_2)}{(x-x_0)(x-x_2)}y_1 + \frac{(x_2-x_0)(x_2-x_1)}{(x-x_0)(x-x_1)}y_2$
- $y = f(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_2$
- $y = f(x) = \frac{(x-x_1)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_0 + \frac{(x-x_0)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_2$

MTH603:Quiz no.2

Question # 6 of 10 (Start time: 08:51:03 AM, 30 January 2021)

x: 1 3 7 f(x): 1 4 9 f(3) Can be found using

Select the correct option

○	Newton's forward difference formula
○	Newton's backward difference formula
◉	Lagrange's interpolation formula
○	None of the given choices

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MTH603:Quiz no.2 Quiz Start Time: 08:48 AM, 3

Question # 7 of 10 (Start time: 08:51:23 AM, 30 January 2021)

Given the following data x:1 2 7 11 y:6 10 13 37 Which formula is useful in finding the interpolating polynomial?

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Select the correct option

Reload M

○	Newton's forward difference interpolation formula
○	Newton's backward difference interpolation formula
◉	Lagrange's interpolation formula
○	None

MTH603:Quiz no.2

Quiz Start Time: 08:48 AM, 30 January 2021

Question # 8 of 10 (Start time: 08:52:00 AM, 30 January 2021) **Total Marks:**

If there are (n+2) values of y corresponding to (n+2) values of x, then we can represent the function f(x) by a polynomial of degree

Select the correct option

Reload Math Equations

- ○ n+2
- ◉ n+1
- ○ n
- ○ n-1

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MTH603:Quiz no.2 Quiz Start

Question # 9 of 10 (Start time: 08:52:24 AM, 30 January 2021)

Given the following data x: 0 1 2 4 y: 1 1 2 6 Value of 1st order divided difference f[2,4] is

Select the correct option

- 1
- 2
- 0
- -2

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MTH603:Quiz no.2 Quiz Start Time: 08:48 AM, 30 Janu

Question # 10 of 10 (Start time: 08:52:52 AM, 30 January 2021) Total

For the given data points $(1, 0.3)$, $(3, 1)$, *and* $(5, 1.2)$ *the divide difference table will be given as*

Select the correct option

Reload Math Equ

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x	y	$1stD.D$	$2ndD.D$
2	0.3	0.35	
4	1	0.1	-0.125
6	1.2		

x	y	$1stD.D$	$2ndD.D$
2	0.3	0.35	
4	1	0.1	-0.225
6	1.2		

x	y	$1stD.D$	$2ndD.D$
2	0.3	0.35	
4	1	0.1	-0.525
6	1.2		

x	y	$1stD.D$	$2ndD.D$
2	0.3	0.35	
4	1	0.1	-0.0625
6	1.2		

Thank you For Watching☺

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